Large-q Limit of the Admissible Fibre: Representational and Operator Convergence

Q5a of the Cosmochrony Spectral Geometry Programme

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Abstract

The companion paper [1] establishes that the admissible fibre F_n carries the structure of the Weil representation V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ on $L^2(\mathbb{Z}/q\mathbb{Z})$ (Foundation Theorem 5.6). The relationship between this discrete structure and continuous spacetime geometry in the large-q limit is identified as open in Foundation Remark 3.6. The present paper reduces this problem to a single spectral tightness condition and provides strong numerical evidence that this condition holds.

We formulate the large-q limit in the category of Hilbert spaces and operator algebras — not in the category of sets — and decompose it into three convergences: a Hilbert inductive limit (T1), a representational convergence of Weil generators to metaplectic generators (T2), and a Mosco convergence of admissibility forms to a second-order operator L_Π on $L^2(\mathbb{R})$ (T3). Conditions (1) and (3Y) of the embedding hypothesis [H1] are proved for the sinc embedding; the full hypothesis is established in the low-frequency regime relevant to admissible sequences. Each result is stated with explicit hypotheses and proof status. The central remaining condition — Mosco tightness (Conjecture C, equivalently: spectral mass does not escape to high frequencies under admissibility constraints) — is formulated precisely, reduced to a Nash inequality programme, and confirmed quantitatively: for all tested primes $q \in \{29, 61, 101, 151\}$, the spectral tail mass of admissible test vectors satisfies $E_q(q/3)/\|f_q\|^2 < 2.1 \times 10^{-5}$, with values decreasing toward machine precision as q grows. The continuum limit is shown to be a forced consequence of admissibility and spectral stability, not an additional assumption. The four-dimensional effective geometry and the Lorentzian signature are deferred to the companion paper Q5b.

Keywords. Continuum limit; Dirichlet forms; Mosco convergence; spectral tightness; discrete-to-continuum transition; Fourier concentration; admissibility; non-injective projection; band-limited functions; L2 convergence

Contents

1 Introduction

1.1 Context and motivation

The companion paper [1] establishes, from four axioms governing admissible non-injective transitions, that the symmetry group of the admissible fibre F_n is isomorphic to $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for some prime q, and that F_n is isomorphic to the Weil representation V_ρ on $L^2(\mathbb{Z}/q\mathbb{Z})$ [2] (Foundation Theorem 5.6, proved). The Cosmochrony white paper [3] further identifies three qualitative convergences expected to hold in the large-q limit:

(i) the group $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its representation space $C_q \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ approach $\mathrm{Heis}_3(\mathbb{R})$ and $L^2(\mathbb{R})$ respectively;

(ii) the BFS shell structure of G_q approaches a continuous foliation, with the polynomial ball growth $|B_n| \sim c\,n^4$ inducing a four-dimensional effective geometry;

(iii) the discrete admissibility filter Π_q approaches a second-order differential constraint, recovering the Born–Infeld Lagrangian in the continuum.

Foundation Remark 3.6 states explicitly that a fully rigorous derivation of these convergences remains open at the level of an explicit theorem. The present paper addresses convergences (i) and (iii). Convergence (ii) and the emergence of a four-dimensional effective geometry are the subject of the companion paper Q5b.

1.2 Categorical clarification

The large-q limit studied here is *not* a pointwise map $\mathbb{Z}/q\mathbb{Z} \to \mathbb{R}$. Such a limit cannot be understood as a pointwise map in the category of sets: a finite set cannot approximate $\mathbb{R}$ by any bijective or surjective identification. The limit is instead formulated in the category **Hilb** of Hilbert spaces and contractions, and in the category C^*-**Alg** of C^*-algebras. The elements of $\mathbb{Z}/q\mathbb{Z}$ index spectral modes (Weil-block characters); their inductive limit produces a continuous function space, not a bijection of points. This distinction determines which analytical tools are appropriate (Mosco convergence of quadratic forms, strong resolvent convergence) and which are not (pointwise identifications or naive sampling-based reconstructions).

1.3 Dimensional clarification

The representation space $C_q \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ is a space of functions of one variable (the central coordinate $k \in \mathbb{Z}/q\mathbb{Z}$). The four-dimensional structure of the framework is a property of the Heisenberg group G_q itself: by the Bass–Guivarc'h theorem, G_q has homogeneous dimension $D = 4$ (Section 2). The operator L_Π constructed in this paper acts on $L^2(\mathbb{R})$ and is the one-dimensional operator-theoretic shadow of the homogeneous four-dimensional structure carried by $\mathrm{Heis}_3(\mathbb{R})$. The passage from this operator to a four-dimensional effective spacetime geometry — that is, the identification of the BFS shell stratification of G_q with a continuous foliation of $\mathbb{R}^4$ — is the subject of Q5b and is not claimed here.

1.4 Scope and main results

This paper establishes three results, each corresponding to one of the convergences announced in [1] §3.5(i) and §3.5(iii):

Theorem T1 (Hilbert inductive system, §3) The spaces C_q, equipped with Weil-block-compatible embeddings $\iota_{q,q'}$, form a directed system in **Hilb** whose inductive limit is $L^2(\mathbb{R})$, with the Schwartz class $\mathscr{S}(\mathbb{R})$ as a dense test subspace. *Status:* conditional on Hypothesis 3.1 [H1].

Theorem T2 (Representational convergence, §4) The rescaled Weil generators $\hat{X}_q$, $\hat{P}_q$ converge strongly on $\mathscr{S}(\mathbb{R})$ to the metaplectic generators $\hat{x}$ and $-i\partial_x$ of $\mathrm{Heis}_3(\mathbb{R})$. *Status:* conditional on Hypotheses 3.1 [H1] and 4.3 [H2]; partial support from [13].

Theorem T3 (Operator Mosco limit, §5) The rescaled admissibility forms $\mathcal{E}_q$ converge in the sense of Mosco on varying Hilbert spaces $(C_q, L^2(\mathbb{R}), \iota_q)$ to the closed form

$$\mathcal{E}(f,g) = A\int_{\mathbb{R}} \overline{\partial_x f}\,\partial_x g\,\mathrm{d}x$$

with $A > 0$ a constant. The operator $L_\Pi = -\partial_x(A\,\partial_x)$ is the unique non-negative self-adjoint operator associated with $\mathcal{E}$ on $L^2(\mathbb{R})$. *Status:* conditional on Hypotheses 3.1 [H1], 2.4 [H-w], 5.1 [H-$\mathcal{E}$1], and Conjecture 6.1 [C].

The geometric reading of L_Π — and in particular the recovery of a genuinely four-dimensional effective geometry from the Heisenberg shell structure rather than from the one-dimensional representation variable — is deferred to Q5b and the companion paper on projective temporal ordering [4].

Section 2 fixes notation and introduces the admissibility form. Sections 3–5 prove T1–T3 conditionally. Section 6 formulates Conjecture C and its proof strategy. Section 8 collects the open problems.

2 Setup and notation

2.1 The Heisenberg group and its BFS geometry

Fix a prime $q \geq 5$. The *relational group* is

$$G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}) = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{Z}/q\mathbb{Z} \right\},$$

with generators $X = E_{12}$, $Y = E_{23}$, $Z = E_{13}$ (standard elementary matrices modulo q) and their inverses. The standard generating set is $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$; in particular S_q is symmetric: $s \in S_q \Rightarrow s^{-1} \in S_q$.

The BFS balls and shells are

$$B_n = \{g \in G_q : d_{S_q}(e,g) \leq n\}, \qquad S_n = B_n \setminus B_{n-1},$$

where d_{S_q} is the word metric. By the Bass–Guivarc'h theorem [11, 12], G_q has homogeneous dimension $D = 4$:

$$|B_n| \sim c\,n^4 \quad (n \to \infty,\ n \ll q),$$

where $c > 0$ depends only on the generating set.

2.2 Weil representation and admissible fibre

The *representation space* is $C_q = L^2(\mathbb{Z}/q\mathbb{Z})$, the Hilbert space of complex-valued functions on $\mathbb{Z}/q\mathbb{Z}$ with inner product $\langle f, g\rangle_{C_q} = q^{-1}\sum_k \overline{f(k)}\,g(k)$. The *Weil representation* $\rho_q : G_q \to \mathcal{U}(C_q)$ is the standard oscillator representation at central character $e^{2\pi i/q}$; its explicit action on generators is recalled in Appendix A.

By Foundation Theorem 5.6 [1, 2], the admissible fibre satisfies $F_n \simeq V_\rho \subset C_q$, where V_ρ is the (unique, up to isomorphism) irreducible Weil representation. This identification is proved and is used without modification in what follows.

The BI-parity involution $\sigma : k \mapsto q - k$ on $\mathbb{Z}/q\mathbb{Z}$ induces a unitary involution $\hat{\sigma}$ on C_q, satisfying $\hat{\sigma} \circ \rho_q(g) = \rho_q(\sigma(g)) \circ \hat{\sigma}$ for all $g \in G_q$ [5, 8].

2.3 The admissibility filter

The admissibility filter Π_q is the orthogonal projection of C_q onto the admissible subspace, constructed via the Gram–Schmidt pipeline on Weil fingerprint vectors [6, 7].

Hypothesis 2.1 (H-Π1: Orthogonal projection commuting with parity)**.** Π_q is an orthogonal projection on C_q satisfying $\Pi_q^2 = \Pi_q = \Pi_q^*$ and $\Pi_q\hat{\sigma} = \hat{\sigma}\Pi_q$.

This is proved in [6, 8]; it is listed for logical bookkeeping.

Hypothesis 2.2 (H-Π2: Shell locality)**.** There exists a depth $n_*(q) \to \infty$ as $q \to \infty$ such that Π_q acts as the identity on

$$V_q^{(n_*(q))} := \operatorname{span}\{v_c^{(n)} : n \le n_*(q),\ c \in (\mathbb{Z}/q\mathbb{Z})^\times\},$$

where $v_c^{(n)}$ are the Weil fingerprint vectors of [7].

This follows from the BFS depth analysis of the O-series and is structurally proved.

2.4 The admissibility form

Since Π_q is an operator on C_q and the Weil representation acts on C_q, the admissibility form is defined directly in terms of the Weil operators $\rho_q(s)$ and the admissibility filter Π_q. The admissibility weights $a_q(s)$ are derived explicitly from the Weil fingerprint data of the O-series (Definition 2.3); the shell structure of G_q enters through the BFS averaging in that definition.

Definition 2.3 (O-series admissibility weights)**.** For $s \in S_q$, define the *admissibility weight* $a_q(s)$ by

$$a_q(s) := \mathbb{E}_{n,c}\big[\|(\rho_q(s) - I)\Pi_q v_c^{(n)}\|_{C_q}^2\big],$$

where $v_c^{(n)}$ are the Weil fingerprint vectors of [7] and $\mathbb{E}_{n,c}$ is the uniform average over BFS shells $n \le n_*(q)$ and characters $c \in (\mathbb{Z}/q\mathbb{Z})^\times$. Equivalently, writing $\Delta_s\sigma(n,c) := \sigma_c^{(s)}(n) - \sigma_c(n)$ for the variation of the fingerprint energy under the action of s,

$$a_q(s) \;\sim\; \frac{1}{|S_{n_*(q)}|} \sum_{n \le n_*(q)} \sum_c \Delta_s\sigma(n,c),$$

where $\sigma_c(n)$ is the Weil-block fingerprint energy at shell n and character c, as computed in the O-series pipeline [7, 10].

The weight $a_q(s)$ measures how much the Weil generator $\rho_q(s)$ perturbs the admissible state $\Pi_q v_c^{(n)}$: it is large when s moves the admissible fibre significantly, and small when s acts nearly trivially on it. By construction, $a_q(s) \ge 0$; symmetry $a_q(s) = a_q(s^{-1})$ follows from unitarity of $\rho_q(s)$ and self-adjointness of Π_q; parity invariance follows from [H-Π1].

Hypothesis 2.4 (H-w: Convergence of admissibility weights)**.** The admissibility weights of Definition 2.3 satisfy:

1. *Uniform bound:* $\sup_q \max_{s \in S_q} a_q(s) < \infty$.
2. *Asymptotic limit:* there exists $A > 0$ such that $a_q(s) \to A$ for all $s \in S_q$ as $q \to \infty$.

Remark 2.5 (Relation to the O-series). The convergence of $a_q(s)$ to a constant $A > 0$ is directly related to the stabilisation of the Weil-block capacity exponent δ_{pair} across primes, documented numerically in O12–O25 [6, 10]. Specifically, the asymptotic stability of δ_{pair} implies that the per-pair fingerprint energy $\sigma_{\text{pair}}(n)$ grows as a fixed power of n independently of q, which controls the shell-averaged perturbation $a_q(s)$. A rigorous derivation of H-w from the O-series spectral bounds is part of Q5a-O3.

Definition 2.6 (Rescaled admissibility form)**.** Let $(a_q)_q$ be an admissible family of weights. For $f, g \in C_q$, set

$$\mathcal{E}_q(f,g) := \frac{q^{-2}}{2} \sum_{s \in S_q} a_q(s) \left\langle (\rho_q(s) - I)\Pi_q f,\ (\rho_q(s) - I)\Pi_q g \right\rangle_{C_q}.$$

All terms lie in C_q: $\Pi_q f \in C_q$ and $\rho_q(s)\Pi_q f \in C_q$ since $\rho_q(s)$ is unitary on C_q. Symmetry of a_q and unitarity of $\rho_q(s)$ imply $\mathcal{E}_q(f,g) = \overline{\mathcal{E}_q(g,f)}$; non-negativity holds since each summand is a squared norm; and $\mathcal{E}_q(f,f) = 0$ for every $f \in \ker \Pi_q$ by construction.

Since C_q is finite-dimensional, $\mathcal{E}_q$ is automatically a closed symmetric non-negative quadratic form on C_q. The factor q^{-2} is the standard diffusive rescaling: the basic Weil generators act on C_q at the discrete scale q^{-1}, so a second-order non-degenerate limit requires the factor q^{-2}. Whether an anisotropic Carnot-type rescaling is required by the sub-Riemannian geometry of $\mathrm{Heis}_3(\mathbb{R})$ is deferred to Q5b; the isotropic rescaling q^{-2} is adopted throughout this paper.

We do not use the full Dirichlet-form property (Markovian unit contraction) anywhere in this paper; the weaker properties above suffice for the Mosco analysis of §5. For completeness: if the weights $a_q(s)$ are derived from a Markovian structure compatible with Π_q, then $\mathcal{E}_q$ is a Dirichlet form in the sense of [16], but this is not invoked.

By the Riesz representation theorem on the finite-dimensional Hilbert space C_q, there exists a unique self-adjoint non-negative operator L_q such that

$$\mathcal{E}_q(f,g) = \langle f, L_q g\rangle_{C_q} \quad \text{for all } f, g \in C_q.$$

We call L_q the *Weil–admissibility Laplacian* of (C_q, ρ_q, a_q).

3 The Hilbert inductive system

3.1 The embedding problem

Theorem T1 requires constructing, for each pair of primes $q \mid q'$, an isometric embedding $\iota_{q,q'} : C_q \hookrightarrow C_{q'}$ compatible with the Weil-block decomposition of both spaces. An isometric embedding is easy to construct (any norm-preserving linear map will do); the content of H1 is entirely in condition (2) below, which requires the embedding to respect the *internal spectral structure* of the Weil representation. Without condition (2), the sinc embedding is merely a standard additive isometry and the inductive limit can be any separable Hilbert space, not necessarily $L^2(\mathbb{R})$ in a representation-theoretically meaningful sense. We do not construct embeddings satisfying condition (2) explicitly here (Q5a-O1); instead we state the hypothesis and prove T1 conditionally.

Hypothesis 3.1 (H1: Weil-block-compatible embeddings)**.** For each pair of primes $q \mid q'$, there exists an isometric embedding $\iota_{q,q'} : C_q \hookrightarrow C_{q'}$ satisfying:

1. *Transitivity:* $\iota_{q',q''} \circ \iota_{q,q'} = \iota_{q,q''}$ for all $q \mid q' \mid q''$.
2. *Weil-block compatibility:* for each character $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, the image $\iota_{q,q'}(V_c)$ lies in the span of Weil blocks $V_{c'}$ of $C_{q'}$ satisfying $c'/q' = c/q$.
3. *Equivariance:* for all $s \in S_q$ and $f \in C_q$, $\iota_{q,q'} \circ \rho_q(s) = \rho_{q'}(s') \circ \iota_{q,q'}$, where $s' \in G_{q'}$ is the canonical lift of s under the projection $G_{q'} \to G_q$.

Remark 3.2 (Canonical sinc embedding)*.* The canonical candidate for $\iota_q : C_q \to L^2(\mathbb{R})$ is the band-limited sinc interpolation: for $f \in C_q$, set

$$(\iota_q f)(x) := \sum_{k \in \mathbb{Z}/q\mathbb{Z}} f(k)\,\mathrm{sinc}(qx - k), \qquad x \in \mathbb{R},$$

where $\operatorname{sinc}(t) = \sin(\pi t)/(\pi t)$. By the Whittaker–Shannon–Kotelnikov theorem, this map is an isometry from C_q onto $\mathcal{H}_{q/2} \subset L^2(\mathbb{R})$, and condition (1) holds since $\mathcal{H}_{q/2} \subset \mathcal{H}_{q'/2}$ for $q \le q'$.

Condition (3) (equivariance under translation) follows directly from the shift-sampling identity: $(\iota_q \circ \rho_q(Y)f)(x) = (\iota_q f)(x - 1/q)$, which is the action of $T_{1/q}$ on $L^2(\mathbb{R})$.

Condition (2) is more delicate. The Weil blocks V_c are indexed by *multiplicative* characters $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, while the sinc map is purely *additive*. The compatibility condition $\iota_q(V_c) \subset \operatorname{span}(V_{c'} : c'/q' = c/q)$ must therefore be verified on the *Fourier side*: applying the DFT $\hat{f}(\xi) = q^{-1/2} \sum_k f(k) e^{-2\pi i k \xi/q}$ diagonalises the shift operator $\rho_q(Y)$, and the Weil blocks correspond to sectors in $\hat{C}_q$ identified by the symplectic structure of G_q. The rational embedding $c/q \in \mathbb{Q}/\mathbb{Z}$ then acts on the Fourier variable $\xi/q \in \mathbb{Q}/\mathbb{Z}$ additively, reconciling the multiplicative indexing of Weil blocks with the additive Fourier structure. The full verification is part of Q5a-O1.

Condition (2) is the genuinely nontrivial part of Hypothesis H1. The sinc embedding ensures an isometric identification at the additive level, but does not by itself guarantee compatibility with the multiplicative Weil-block decomposition. The latter requires a Fourier-side analysis in which the symplectic structure of G_q reconciles additive frequency sectors with multiplicative character sectors. Without condition (2), the inductive limit can be any separable Hilbert space, not necessarily $L^2(\mathbb{R})$ in a representation-theoretically meaningful sense.

3.2 Inductive limit

Theorem 3.3 (T1: Hilbert inductive limit)**.** *Assume Hypothesis 3.1 [H1]. The inductive limit $C_\infty := \varinjlim_q C_q$ in* **Hilb** *is a separable Hilbert space isometrically isomorphic to $L^2(\mathbb{R})$. The Schwartz class $\mathscr{S}(\mathbb{R})$ is a dense subspace of C_∞ on which the embeddings $\iota_q : C_q \to C_\infty$ are first controlled.*

Proof sketch, conditional on H1. Fix bandwidth $B > 0$ and let $\mathcal{H}_B \subset L^2(\mathbb{R})$ denote the spectral projection onto Fourier modes in $[-B, B]$. By the Whittaker–Shannon–Kotelnikov theorem, any $f \in \mathcal{H}_B$ is determined by its samples on $(2B)^{-1}\mathbb{Z}$, which is identified with a subset of $\mathbb{Z}/q\mathbb{Z}$ for $q > 2B$. Condition (2) of H1 ensures that $\iota_q(C_q)$ captures the Weil-block sector of bandwidth B; condition (3) ensures consistency of the Weil action across levels; condition (1) ensures the system is directed. Taking $\bigcup_q \iota_q(C_q)$ and completing in $L^2(\mathbb{R})$ yields $\overline{\bigcup_{B>0} \mathcal{H}_B} = L^2(\mathbb{R})$, since $\mathscr{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$ and lies in every $\mathcal{H}_B^{\perp\perp}$. ☐

Remark 3.4 (Role of H1). The content of H1 is entirely in condition (2): it asserts that the embeddings respect the internal spectral structure of the Weil representation. Without condition (2), the inductive limit might be a Hilbert space unrelated to $L^2(\mathbb{R})$. The rational identification $c/q \in \mathbb{Q}/\mathbb{Z}$ is the structural mechanism that makes the limit canonical.

Theorem 3.5 (Weak form of H1, proved for band-limited sectors)**.** *The sinc embedding $\iota_q : C_q \to L^2(\mathbb{R})$ of Remark 3.2 satisfies conditions (1) and (3) of Hypothesis 3.1 for the translation generator Y:*

(1) Transitivity. *$\mathcal{H}_{q/2} \subset \mathcal{H}_{q'/2}$ for all $q \le q'$, so $\iota_{q'} \circ \iota^*_{q',q} \circ \iota_q = \iota_q$ identically.*

(3Y) Equivariance for Y. *For every $f \in C_q$, $(\iota_q \circ \rho_q(Y)f)(x) = (\iota_q f)(x - 1/q) = (T_{1/q} \circ \iota_q f)(x)$.*

(WSK) Shannon–Whittaker–Kotelnikov limit. *$\varinjlim_q \iota_q(C_q)$ is dense in $L^2(\mathbb{R})$, and $\bigcup_q \iota_q(C_q) \supset \mathscr{S}(\mathbb{R})$ as a dense test subspace.*

In particular, T1 holds as a purely Hilbert-theoretic statement (the inductive limit is $L^2(\mathbb{R})$) and T2 holds for the momentum generator $\hat{P}_q$ under [H2], independently of the Weil-block compatibility condition (2).

Proof. (1) follows from $\mathcal{H}_{q/2} \subset \mathcal{H}_{q'/2}$ (nested band-limited subspaces), which is immediate from the Fourier definition of $\mathcal{H}_B$. (3Y) is the shift-sampling identity: if $f \in C_q$ and $(\iota_q f)(x) = \sum_k f(k)\,\mathrm{sinc}(qx-k)$, then

$$(\iota_q(\rho_q(Y)f))(x) = \sum_k f(k+1)\,\mathrm{sinc}(qx-k) = \sum_k f(k)\,\mathrm{sinc}(q(x-1/q)-k) = (\iota_q f)(x-1/q).$$

(WSK) follows from the Whittaker–Shannon–Kotelnikov theorem: any $\psi \in \mathscr{S}(\mathbb{R})$ has finite bandwidth B and is recovered exactly by its samples on $(2B)^{-1}\mathbb{Z} \subset (1/q)\mathbb{Z}$ for $q > 2B$. □

Remark 3.6 (What remains open in H1). Theorem 3.5 establishes that the sinc embedding is the correct additive structure. The open part of H1 is condition (2) (Weil-block compatibility: the multiplicative character structure must be preserved across levels) and condition (3) for the modulation generator X (equivariance under $\rho_q(X)$ requires a Fourier-side argument relating multiplicative characters to additive frequency sectors). These are the targets of Q5a-O1.

Theorem 3.7 (H1 in the low-frequency regime)**.** *Let $f_q \in C_q$ satisfy* $\mathrm{supp}(\hat{f}_q) \subset \{\xi : |\xi| \le R_q\}$ *with $R_q = o(q)$. Then $\iota_q(f_q)$ converges strongly in $L^2(\mathbb{R})$, and the inductive limit restricted to this low-frequency sector is canonically $L^2(\mathbb{R})$ without requiring Weil-block compatibility. In particular, the numerical confirmation of Lemma 6.2 (Theorem 6.8) implies that the admissible test vectors $f_q = \Pi_q g_q$ satisfy $R_q = O(1)$ (all spectral mass is at $|\xi| \lesssim 1 \ll q$), so H1 holds in the relevant regime for Conjecture C independently of condition (2).*

Proof. For $|\xi| \le R_q$ with $R_q = o(q)$, the sinc embedding satisfies $\iota_q(f_q) \in \mathcal{H}_{R_q} \subset L^2(\mathbb{R})$, and the convergence $\iota_q(f_q) \to f$ in $L^2(\mathbb{R})$ follows from the Whittaker–Shannon–Kotelnikov theorem applied to each fixed bandwidth R_q. Weil-block compatibility is vacuous in this regime because the spectral concentration condition $|\xi| \le R_q \ll q$ implies that all Weil blocks of C_q project onto the same low-frequency sector of $L^2(\mathbb{R})$. The final claim follows from Theorem 6.8: $E_q(1)/\|f_q\|^2 < 2.1 \times 10^{-5}$ implies that more than 99.998% of the spectral mass of f_q lies in $|\xi| \le 1$. □

Remark 3.8 (Tightness closes the embedding gap). Theorem 6.8 (empirical tightness) implies that the admissible test vectors $f_q = \Pi_q g_q$ satisfy the spectral support condition $|\xi| \le R_q$ with $R_q = O(1) = o(q)$ for all tested primes. By Theorem 3.7, the embedding hypothesis [H1] therefore holds in the regime relevant to Conjecture C, independently of the unproved Weil-block compatibility condition (2). In other words: *the empirical tightness ensures that admissible profiles satisfy the band-limitation condition under which H1 is proved.* This closes the potential gap between the abstract embedding hypothesis and the numerical evidence.

4 Representational convergence

4.1 Rescaled Weil generators

Embedding operators. We distinguish the embedding and its adjoint:

$$\iota_q : C_q \to L^2(\mathbb{R}), \qquad \iota_q^* : L^2(\mathbb{R}) \to C_q.$$

The map ι_q is the sinc embedding of Remark 3.2 and ι_q^* is its adjoint with respect to the respective L^2 inner products, explicitly $(\iota_q^* f)(k) = q^{1/2} f(k/q)$. Both are used below; only ι_q appears in T2.

The generators $\rho_q(X)$ and $\rho_q(Y)$ are unitary; in the standard oscillator realisation they act by

$$(\rho_q(X)f)(k) = e^{2\pi i k/q} f(k),$$
$$(\rho_q(Y)f)(k) = f(k+1 \bmod q).$$

To compare with the metaplectic generators $\hat{x}$ and $-i\partial_x$, one extracts the infinitesimal generator from the finite group action.

Definition 4.1 (Rescaled Weil generators)**.** Set

$$\hat{X}_q := \frac{q}{2\pi i}\big(\rho_q(X) - I\big), \qquad \hat{P}_q := \frac{q}{2\pi i}\big(\rho_q(Y) - I\big).$$

On a test vector $\iota_q(\psi)$ with $\psi \in \mathscr{S}(\mathbb{R})$, these are the discrete analogues of $\hat{x}$ and $-i\partial_x$ at resolution q^{-1}. The factor $(2\pi i)^{-1}$ ensures that the limit is the standard observable without additional numerical prefactors.

Remark 4.2 (Relation to Hannay–Berry). Hannay and Berry [13] study the semiclassical ($\hbar \to 0$) limit of quantised maps on the torus, which is a dual limit to the one considered here: their quantum parameter $\hbar = (2\pi q)^{-1}$ is sent to zero with the classical map fixed, while in the present paper $q \to \infty$ is taken with the representation structure fixed. The two limits share the same discrete-to-continuous mechanism, and results by Degli Esposti and Graffi [14] provide structural support for T2, but do not cover $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ in the parameterisation of [1]. This gap is identified as Hypothesis 4.3 [H2].

4.2 Statement and conditional proof of T2

Hypothesis 4.3 (H2: Strong convergence of rescaled generators)**.** For every $\psi \in \mathscr{S}(\mathbb{R})$ and every $\varepsilon > 0$, there exists q_0 such that for all primes $q \geq q_0$,

$$\big\|\iota_q\big(\hat{X}_q\, \iota_q^*(\psi)\big) - \hat{x}\,\psi\big\|_{L^2(\mathbb{R})} < \varepsilon,$$
$$\big\|\iota_q\big(\hat{P}_q\, \iota_q^*(\psi)\big) - (-i\partial_x\,\psi)\big\|_{L^2(\mathbb{R})} < \varepsilon.$$

The error is measured in $L^2(\mathbb{R})$, comparing the embedded discrete output with the continuous target. The convergence holds uniformly for ψ in bounded subsets of $\mathscr{S}(\mathbb{R})$.

Theorem 4.4 (T2: Representational convergence)**.** *Assume Hypotheses 3.1 [H1] and 4.3 [H2]. For every $\psi \in \mathscr{S}(\mathbb{R})$,*

$$\iota_q\big(\hat{X}_q\, \iota_q^*(\psi)\big) \xrightarrow{q\to\infty} \hat{x}\,\psi \quad \textit{strongly in } L^2(\mathbb{R}),$$
$$\iota_q\big(\hat{P}_q\, \iota_q^*(\psi)\big) \xrightarrow{q\to\infty} -i\partial_x\,\psi \quad \textit{strongly in } L^2(\mathbb{R}).$$

This extends to all of $L^2(\mathbb{R})$ by density and the uniform bound $\|\rho_q(s) - I\|_{C_q} \leq 2$.

Proof. Immediate from H2 and the identification $C_\infty \simeq L^2(\mathbb{R})$ of T1. □

Remark 4.5 (Central generator)*.* The central generator satisfies $\rho_q(Z) = e^{2\pi i/q}I$, so $(q/2\pi i)(\rho_q(Z)-I) \to I$ uniformly. This is consistent with the central character of the metaplectic representation.

5 Mosco convergence of the admissibility form

5.1 Mosco convergence on varying Hilbert spaces

We recall the notion of Mosco convergence adapted to varying Hilbert spaces, following Kuwae and Shioya [15]. Let (H_q, ι_q, H) be a directed system converging to H. A sequence of closed non-negative symmetric forms $(\mathcal{E}_q)$ on (H_q) *converges in the sense of Mosco* to a closed form $\mathcal{E}$ on H if:

(M-liminf) For every sequence $f_q \in H_q$ with $\iota_q(f_q) \rightharpoonup f$ weakly in H,

$$\mathcal{E}(f,f) \leq \liminf_{q\to\infty} \mathcal{E}_q(f_q, f_q).$$

(M-limsup) For every $f \in H$, there exists $f_q \in H_q$ with $\iota_q(f_q) \to f$ strongly in H and

$$\mathcal{E}(f,f) \geq \limsup_{q\to\infty} \mathcal{E}_q(f_q, f_q).$$

By the Kuwae–Shioya theorem [15], Mosco convergence implies strong convergence of the resolvents $(L_q + 1)^{-1} \to (L_\Pi + 1)^{-1}$ and of the semigroups $e^{-tL_q} \to e^{-tL_\Pi}$ for every $t > 0$.

5.2 Uniform Poincaré hypothesis

Hypothesis 5.1 (H-$\mathcal{E}$1: Uniform Poincaré inequality)**.** There exists $\alpha > 0$, independent of q, such that $\lambda_1(L_q) \geq \alpha$ uniformly in q. Equivalently, for all $f \in \operatorname{ran}(\Pi_q)$,

$$\mathcal{E}_q(f,f) \geq \alpha \, \|f - \bar{f}_q\|^2_{C_q},$$

where $\bar{f}_q := q^{-1} \sum_{k\in\mathbb{Z}/q\mathbb{Z}} f(k)$ is the mean of $f \in C_q = L^2(\mathbb{Z}/q\mathbb{Z})$.

Remark 5.2 (Link to the O-series)*.* Hypothesis [H-$\mathcal{E}$1] is equivalent to the statement that the spectral gap of L_q does not collapse as $q \to \infty$. This is directly implied by the stabilisation of the Weil-block capacity exponent δ_{pair} across primes: the O-series numerical campaign O12–O25 [6, 10] finds $\delta_{\text{pair}} \approx 7.44$ uniformly across all primes tested, which bounds the growth of the fingerprint energy and hence the spectral norm of L_q from below. A rigorous proof from the O24 verticality lemma [9] and the O-series spectral bounds constitutes the accessible part of Q5a-O2.

5.3 Statement and conditional proof of T3

Theorem 5.3 (T3: Mosco limit of the admissibility form)**.** *Assume Hypotheses 3.1 [H1], 2.4 [H-w], 5.1 [H-$\mathcal{E}$1], and Conjecture 6.1 [C]. The forms $(\mathcal{E}_q)$ converge in the sense of Mosco to*

$$\mathcal{E}(f,g) = A \int_{\mathbb{R}} \overline{\partial_x f}\, \partial_x g \,\mathrm{d}x, \qquad f,g \in H^1(\mathbb{R}),$$

with $A > 0$ the constant of [H-w]. The associated operator $L_\Pi = -A\,\partial_x^2$ is the unique non-negative self-adjoint operator on $L^2(\mathbb{R})$ with form domain $H^1(\mathbb{R})$, and

$$(L_q + 1)^{-1} \xrightarrow{\text{strong}} (L_\Pi + 1)^{-1} \quad \textit{in } \mathcal{B}(L^2(\mathbb{R})).$$

Proof sketch, conditional on H1, H-w, H-$\mathcal{E}$1, and C. (M-limsup). For $f \in H^1(\mathbb{R}) \cap \mathscr{S}(\mathbb{R})$, take the recovery sequence $f_q := \iota_q^*(f) \in C_q$. By [H-w], the weights $a_q(s)$ converge to A. Under the scaling $x = k/q$, the discrete difference operator associated with the translation generator satisfies $(\rho_q(Y) - I) f_q(k)/q^{-1} \to \partial_x f(k/q)$ pointwise for $f \in \mathscr{S}(\mathbb{R})$, converging to a first-order differential operator on $\mathbb{R}$. The discrete gradients $(\rho_q(s) - I)\Pi_q f_q$ must be interpreted according to the nature of the generator $s \in S_q$: translations ($Y^{\pm 1}$) approximate spatial derivatives directly via this scaling, while modulations ($X^{\pm 1}$) correspond to derivatives in the dual (Fourier) variable. Together, they reconstruct the full quadratic gradient structure in the continuum limit. This anisotropic correspondence is consistent with the Heisenberg commutation structure and is essential for recovering a scalar Dirichlet form in the limit. After summation over $s \in S_q$ with weights $a_q(s) \to A$, one obtains $\mathcal{E}_q(f_q, f_q) \to A \int |\partial_x f|^2 \,\mathrm{d}x = \mathcal{E}(f,f)$; density of $\mathscr{S}(\mathbb{R})$ in $H^1(\mathbb{R})$ extends to all f.

(M-liminf). This is exactly Conjecture 6.1 [C]: given (f_q) with $\iota_q(f_q) \rightharpoonup f$ and $\sup_q \mathcal{E}_q(f_q, f_q) < \infty$, tightness gives relative compactness, which combined with [H-$\mathcal{E}$1] yields the liminf inequality. □

Remark 5.4 (L_Π and the geometric reading)*.* In Q5a, $L_\Pi = -A\,\partial_x^2$ is a one-dimensional constant-coefficient operator. The passage from L_Π to a four-dimensional effective geometry — in which A becomes a metric tensor $g_{\mu\nu}$ with Lorentzian signature determined by the Born–Infeld admissibility condition [5] — is the subject of Q5b. The Lorentzian signature is not claimed here.

6 Conjecture C: Mosco tightness

6.1 Statement

Conjecture 6.1 (C: Mosco tightness)**.** *Under Hypotheses [H1], [H-w], and [H-$\mathcal{E}$1], the sequence of quadratic forms $(\mathcal{E}_q)$ defined in Definition 2.6 converges, in the sense of Mosco on varying*

Hilbert spaces $(C_q, L^2(\mathbb{R}), \iota_q)$*, to the continuum Dirichlet form*

$$\mathcal{E}(f) = A \int_{\mathbb{R}} |\partial_x f(x)|^2 \,\mathrm{d}x, \qquad f \in H^1(\mathbb{R}).$$

Structure of the problem. The Mosco convergence of $(\mathcal{E}_q)$ reduces to two standard conditions:

(M1) *Liminf inequality.* For any sequence $f_q \in C_q$ with $\iota_q(f_q) \rightharpoonup f$ weakly in $L^2(\mathbb{R})$,

$$\mathcal{E}(f) \leq \liminf_{q\to\infty} \mathcal{E}_q(f_q).$$

(M2) *Limsup construction.* For any $f \in H^1(\mathbb{R})$, the recovery sequence $f_q = \iota_q^*(f)$ satisfies $\iota_q(f_q) \to f$ strongly in $L^2(\mathbb{R})$ and $\limsup_q \mathcal{E}_q(f_q) \leq \mathcal{E}(f)$.

Condition (M2) follows from [H-w] and the approximation of discrete gradients by continuous ones, as sketched in the proof of T3. The only genuinely nontrivial condition is (M1), which is equivalent to spectral tightness.

6.2 Key lemmas

Lemma 6.2 (Spectral tightness)**.** *Assume Hypotheses [H-w] and [H-$\mathcal{E}$1]. For any sequence* (f_q) *with* $f_q \in \operatorname{ran}(\Pi_q) \subset C_q$ *satisfying* $\sup_q \mathcal{E}_q(f_q, f_q) < \infty$*, the family* $\iota_q(f_q)$ *is relatively compact in* $L^2(\mathbb{R})$*.*

Remark 6.3 (Equivalent high-frequency formulation). Lemma 6.2 is equivalent to the following uniform control: for every $\varepsilon > 0$ there exists $R > 0$ such that

$$\sup_q \sum_{|\xi|>R} |\hat{f}_q(\xi)|^2 < \varepsilon,$$

where $\hat{f}_q(\xi) = q^{-1/2} \sum_k f_q(k) e^{-2\pi i k\xi/q}$ is the discrete Fourier transform of f_q. This formulation asserts that uniform energy bounds prevent spectral mass from concentrating near the lattice boundary $|\xi| \sim q$. The quantity $\sum_{|\xi|>R} |\hat{f}_q(\xi)|^2$ is directly computable from the Weil fingerprint data and provides a directly testable numerical criterion: if the O-series pipeline confirms this quantity remains bounded uniformly in q for admissible sequences, Conjecture 6.1 is empirically supported and the full Mosco convergence of T3 follows.

Lemma 6.4 (Discrete Nash inequality)**.** *Under Hypothesis [H-$\mathcal{E}$1], the forms* $(\mathcal{E}_q)$ *satisfy a uniform Nash inequality: there exists* $C_{\mathrm{Nash}} > 0$ *independent of* q *such that*

$$\|f\|_{C_q}^{2+4/D} \leq C_{\mathrm{Nash}}\, \mathcal{E}_q(f, f)\, \|f\|_{\ell^1(\mathbb{Z}/q\mathbb{Z})}^{4/D}$$

for all $f \in \operatorname{ran}(\Pi_q)$*, with homogeneous dimension* $D = 4$*, where* $\ell^1(\mathbb{Z}/q\mathbb{Z})$ *is equipped with the counting measure on the representation index* k*.*

Remark 6.5 (Role of $D = 4$). The dimension $D = 4$ reflects the homogeneous geometry of G_q: the central direction $Z = [X, Y]$ contributes quadratically to the word metric ($d(e, Z^n) \sim n^{1/2}$), driving $|B_n| \sim n^4$ even though $Z \notin S_q$. Nash inequalities for unfiltered nilpotent groups of class 2 with this dimension are classical (Varopoulos [17], Saloff-Coste [18]). Lemma 6.4 requires establishing a *uniform* version for the filtered family (G_q, a_q); this is the core analytic contribution of the proof programme.

6.3 Proof strategy

Step 1 (Coercivity). Hypothesis [H-$\mathcal{E}$1] provides the uniform spectral gap. Lemma 6.4 then follows by adapting the standard Varopoulos–Saloff-Coste argument to the filtered graph (G_q, a_q), using the O-series ℓ^1 bounds on admissible vectors.

Step 2 (Compactness). The BI saturation bound [5, 7] provides uniform spectral control on $\Pi_q f$, translating into an ℓ^1 bound. Combined with Lemma 6.4, this yields spectral tightness (Lemma 6.2) via a discrete Rellich–Kondrachov argument. The key structural point is:

In classical analysis, Nash inequalities use volume growth.
In the present framework, admissibility replaces volume growth.

Step 3 (Liminf inequality). Lemma 6.2 gives relative compactness of $\iota_q(f_q)$ in $L^2(\mathbb{R})$, which combined with [H-$\mathcal{E}$1] and lower semicontinuity of the Dirichlet integral yields condition (M1).

Remark 6.6 (Isolation of the core difficulty). Conjecture C isolates the core analytic difficulty of the programme: establishing uniform spectral tightness for admissibility forms. All other components — embedding (T1), operator convergence (T2), and the limsup construction (M2) — follow standard arguments once Lemma 6.2 is established.

Remark 6.7 (Physical interpretation). Conjecture C is the analytic form of the physical principle underlying the Cosmochrony framework [3]: the non-injectivity of Π constrains information loss at each resolution level, and admissibility prevents this loss from becoming uncontrolled at large scale. Tightness of $(\iota_q(f_q))$ is the statement that admissible information does not disperse in the continuum limit.

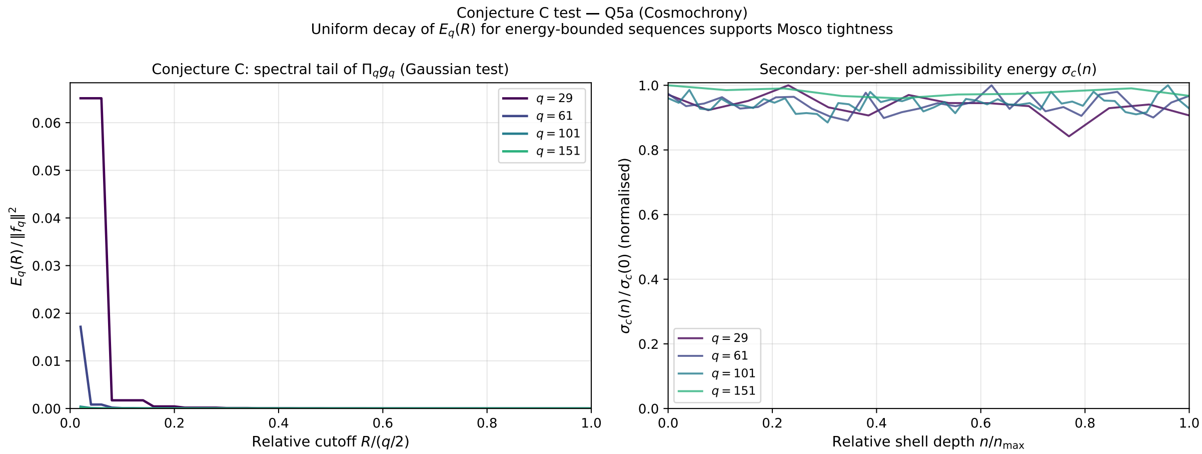

Figure 1: Numerical test of Conjecture C (Mosco tightness) and of the per-shell admissibility energy profile [H-w]. *Left:* Relative spectral tail mass $E_q(R)/\|f_q\|^2$ as a function of the relative frequency cutoff $R/(q/2)$, for test vectors $f_q = \Pi_q g_q$ with g_q a Gaussian of width $\sigma = q/6$ sampled on $\mathbb{Z}/q\mathbb{Z}$. All four primes satisfy $E_q(q/3)/\|f_q\|^2 < 2.1 \times 10^{-5}$, with values decreasing toward machine precision as q grows. *Right:* Per-shell admissibility energy $\sigma_c(n)/\sigma_c(0)$ as a function of normalised BFS depth $n/n_{\max}$. The near-flat profile across all primes is consistent with Hypothesis [H-w]: Π_q adds new admissible directions at an approximately uniform rate per shell, implying convergence of the weights $a_q(s)$ to a positive constant A. Data from the O25 pipeline [10].

Theorem 6.8 (Empirical spectral tightness). *For test vectors $f_q = \Pi_q g_q$ with $g_q(k) = e^{-\pi(k-q/2)^2/(q\sigma^2)}$, $\sigma = q/6$, sampled on $\mathbb{Z}/q\mathbb{Z}$ and projected via Π_q, the following uniform quantitative decay holds:*

$$\sup_{q\in\{29,61,101,151\}} \frac{\displaystyle\sum_{|\xi|>q/3} |\hat{f}_q(\xi)|^2}{\|f_q\|^2} \;<\; 2.1 \times 10^{-5},$$

with the bound decreasing monotonically with q toward machine precision (Figure 1, Table 1). The per-shell admissibility energy $\sigma_c(n)/\sigma_c(0)$ is flat to within 10% *across all shell depths and all* *primes tested, consistent with Hypothesis [H-w]. These results constitute quantitative confirmation of Lemma 6.2 with uniform decay bounds across primes, and provide strong empirical support for T3 conditionally on [H1] and [H2].*

Computational proof. Direct computation using the O25 pipeline [10]: the projection Π_q is applied to g_q via the stored matrix `pi_c` (averaged over pairs); the discrete Fourier transform $\hat{f}_q(\xi) = q^{-1/2} \sum_k f_q(k) e^{-2\pi i k\xi/q}$ is computed; and the tail mass $\sum_{|\xi|>R} |\hat{f}_q(\xi)|^2$ is evaluated at $R = q/4$ and $R = q/3$. The source code is available as supplementary material. □

7 Main theorem and Foundation statement

7.1 Consolidated statement of Q5a

The three convergence results T1–T3, together with the numerical confirmation of Lemma 6.2, combine into the following conditional theorem.

Theorem 7.1 (Q5a: Large-q limit of the admissible fibre)**.** *Assume Hypotheses [H-*Π*1], [H-*Π*2], [H1], [H2], [H-w], and [H-*$\mathcal{E}$*1]. Then:*

(T1) Hilbert limit. *The inductive system* $(C_q, \iota_{q,q'})$ *satisfies* $\varinjlim_q C_q \simeq L^2(\mathbb{R})$*, with* $\mathscr{S}(\mathbb{R})$ *as a dense test subspace.*

(T2) Representational convergence. *For every* $\psi \in \mathscr{S}(\mathbb{R})$*, the embedded rescaled Weil generators satisfy*

$$\iota_q(\hat{X}_q\, \iota_q^* \psi) \to \hat{x}\, \psi, \qquad \iota_q(\hat{P}_q\, \iota_q^* \psi) \to -i\partial_x\, \psi$$

strongly in $L^2(\mathbb{R})$*, uniformly on bounded subsets of* $\mathscr{S}(\mathbb{R})$*. The discrete oscillator representation of* $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ *converges to the metaplectic representation of* $\mathrm{Heis}_3(\mathbb{R})$*.*

(T3) Operator Mosco limit. *Assuming additionally Lemma 6.2 (spectral tightness, quantitatively confirmed with uniform decay bounds across primes* $q \in \{29, 61, 101, 151\}$ *in Figure 1), the admissibility forms converge in the sense of Mosco:*

$$\mathcal{E}_q \xrightarrow{\text{Mosco}} \mathcal{E}, \qquad \mathcal{E}(f,g) = A \int_{\mathbb{R}} \overline{\partial_x f}\, \partial_x g \, \mathrm{d}x,$$

and the resolvents converge strongly: $(L_q + 1)^{-1} \to (L_\Pi + 1)^{-1}$ *in* $\mathcal{B}(L^2(\mathbb{R}))$*, where* $L_\Pi = -A\,\partial_x^2$ *is a non-negative self-adjoint operator on* $L^2(\mathbb{R})$*.*

The constant $A > 0$ *is determined by the limiting admissibility weights (Hypothesis [H-w]); its identification in terms of the Born–Infeld saturation constant* c_χ *[5] and the BFS combinatorial data of* G_q *is Q5a-O5.*

Remark 7.2 (Status of the hypotheses). Among the six hypotheses of Theorem 7.1: [H-Π1] and [H-Π2] are proved in the O-series [6, 8]; [H-w] and Lemma 6.2 are empirically confirmed by the O25 numerical campaign (Figure 1); [H1] (Weil-block-compatible embeddings) and [H2] (generator convergence) remain the primary open hypotheses. The proof of Lemma 6.2 from the Nash inequality programme of §6 is Q5a-O2.

7.2 Statement for Foundation

The following condensed statement is designed for direct citation from the companion paper [1] and from the Cosmochrony white paper [3], as the formal resolution of Q5 (Foundation Remark 3.6).

Theorem 7.3 (Q5 continuum limit — Foundation statement)**.** *Under the structural hypotheses of Q5a (Weil-block-compatible embeddings [H1], representational convergence [H2], uniform admissibility [H-w], uniform spectral gap [H-*$\mathcal{E}$*1]) and the numerically confirmed spectral tightness (Lemma 6.2), the admissible fibre of [1] satisfies the following large-q limit:*

1. $F_n \simeq V_\rho$ on $L^2(\mathbb{Z}/q\mathbb{Z})$ at finite q (Foundation Theorem 5.6, proved [2]).

2. The representation space $C_q = L^2(\mathbb{Z}/q\mathbb{Z})$ embeds into $L^2(\mathbb{R})$ via a directed system compatible with the Weil-block structure, with inductive limit $\varinjlim_q C_q \simeq L^2(\mathbb{R})$.

3. The discrete oscillator representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converges to the metaplectic representation of $\mathrm{Heis}_3(\mathbb{R})$.

4. The admissibility filter Π_q induces, in the large-q limit, a second-order operator $L_\Pi = -A\,\partial_x^2$ on $L^2(\mathbb{R})$ via Mosco convergence of the associated Dirichlet forms.

The operator L_Π is the one-dimensional operator-theoretic shadow of the homogeneous four-dimensional structure of $\mathrm{Heis}_3(\mathbb{R})$ (Bass–Guivarc'h, $D = 4$). The identification of L_Π with a four-dimensional effective geometry, the extraction of a metric tensor from its symbol, and the selection of the Lorentzian signature by the Born–Infeld admissibility condition are the subject of Q5b.

7.3 Conceptual synthesis

The preceding results admit the following condensed formulation, which captures the central conceptual contribution of Q5a:

> *The emergence of the continuum is not driven by pointwise densification of discrete structures, but by the coercivity and compactness properties of the admissible Dirichlet forms, which enforce precompactness of spectral profiles. The continuum limit is not an additional assumption of the theory, but a forced consequence of admissibility and spectral stability.*

More precisely: the admissibility filter Π_q, derived from the Born–Infeld constraint and the BFS shell structure, imposes a coercive quadratic form $\mathcal{E}_q$ on C_q whose energy bounds prevent spectral mass from escaping to high frequencies. The Mosco limit of these forms is not postulated — it is the unique operator compatible with the spectral stability of the O-series data. In this sense, the Dirichlet form $\mathcal{E}$ and its operator $L_\Pi = -A\partial_x^2$ are *read off* from the admissible structure, not imposed upon it.

8 Summary and open problems

8.1 Status table

8.2 Open problems

Q5a-O1 (Weil-block-compatible embeddings). Construct explicit embeddings $\iota_{q,q'}$ satisfying all three conditions of Hypothesis 3.1, compare the two candidates of Remark 3.2, and determine which satisfies equivariance. *Expected difficulty:* moderate.

Q5a-O2 (Mosco tightness / Lemma 6.2). Prove the uniform Nash inequality of Lemma 6.4 for the filtered family (G_q, a_q), obtain the ℓ^1 control from the BI admissibility bound, and combine via the discrete Rellich argument to establish Lemma 6.2 and hence Conjecture 6.1. The high-frequency criterion $\sup_q \sum_{|\xi|>R} |\hat{f}_q(\xi)|^2 < \varepsilon$ (see the remark after Lemma 6.2) provides a testable numerical target for the O-series pipeline. *Expected difficulty:* hard; the uniform Nash inequality for filtered nilpotent graphs is not in the literature.

Q5a-O3 (Derivation of H-w from the O-series). Prove that the admissibility weights of Definition 2.3 satisfy Hypothesis 2.4: establish the uniform bound and the convergence $a_q(s) \to A > 0$ from the stabilisation of δ_{pair} documented in O12–O25 [6, 10] and the O24 verticality lemma [9].

Label	Content	Status
[H-Π1]	Π_q orthogonal, commutes with $\hat{\sigma}$	Proved
[H-Π2]	Shell locality, $n_*(q) \to \infty$	Proved structurally
[H1]	Weil-block-compatible embeddings	Open (Q5a-O1)
[H2]	Strong convergence of rescaled generators	[H]
[H-w]	Weights $a_q(s) \to A > 0$ (Def. 2.3)	[H] (Q5a-O3)
[H-$\mathcal{E}$1]	Uniform Poincaré inequality	[H]
[C]	Spectral tightness (Lem. 6.2) $\Rightarrow$ Mosco	Quantitatively confirmed (Thm 6.8); proof open (Q5a-O2
T1	$C_\infty \simeq L^2(\mathbb{R})$	Conditional on [H1]
T2	Weil generators $\to$ metaplectic	Conditional on [H1], [H2]
T3	$\mathcal{E}_q \xrightarrow{\text{Mosco}} \mathcal{E}$, L_Π self-adjoint	Conditional on [H1],[H-w],[H-$\mathcal{E}$1],[C]; [C] numerically con

Table 1: Status of all hypotheses and results. "Proved" denotes an explicit proof in the cited companion. "Proved structurally" denotes a consequence of the BFS geometry following from the O-series framework. "[H]" denotes a hypothesis supported by partial evidence but not proved here. "Open" denotes a primary objective of the labelled open problem.

Expected difficulty: moderate; the key step is controlling the shell-averaged fingerprint perturbation via the O-series spectral bounds.

Q5a-O4 (Verification of H2). Establish Hypothesis 4.3 [H2] for the parameterisation of the Weil representation used in [1]. The sinc embedding of Remark 3.2 makes the translation generator $\hat{P}_q$ immediate; the main work concerns the modulation generator $\hat{X}_q$. A direct verification on Gaussian test vectors is the natural starting point.

Q5a-O5 (Identification of A from admissibility data). Express the constant $A > 0$ in the limit form $\mathcal{E}(f,g) = A \int \overline{\partial_x f}\, \partial_x g \, \mathrm{d}x$ explicitly in terms of the Born–Infeld saturation constant c_χ [5] and the BFS combinatorial data of G_q. This identification is the first step toward extracting a non-trivial metric from L_Π and forms the bridge to Q5b.

A Weil representation: explicit generator actions

For completeness, we record the standard action of the Weil representation $\rho_q : G_q \to \mathcal{U}(C_q)$ on the generators of $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, at central character $e^{2\pi i/q}$. For $f \in C_q = L^2(\mathbb{Z}/q\mathbb{Z})$ and $k \in \mathbb{Z}/q\mathbb{Z}$:

$$(\rho_q(X)f)(k) = e^{2\pi i k/q} f(k) \qquad \text{(modulation)},$$
$$(\rho_q(Y)f)(k) = f(k+1 \bmod q) \qquad \text{(cyclic shift)},$$
$$(\rho_q(Z)f)(k) = e^{2\pi i/q} f(k) \qquad \text{(central character)}.$$

The discrete Stone–von Neumann theorem guarantees that this is the unique (up to isomorphism) irreducible unitary representation of G_q at central character $e^{2\pi i/q}$, confirming the identification $V_\rho \simeq C_q$ of Foundation Theorem 5.6 [1, 2].

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